

DEPARTMENT OF MATHEMATICS
MAT122 INTRODUCTORY MATHEMATICS II

WEEK 7: LECTURE NOTE

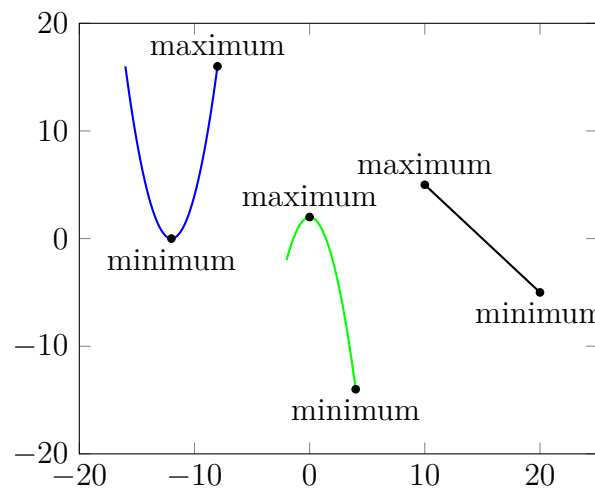
APPLICATIONS OF DIFFERENTIATION

Extrema on an Interval

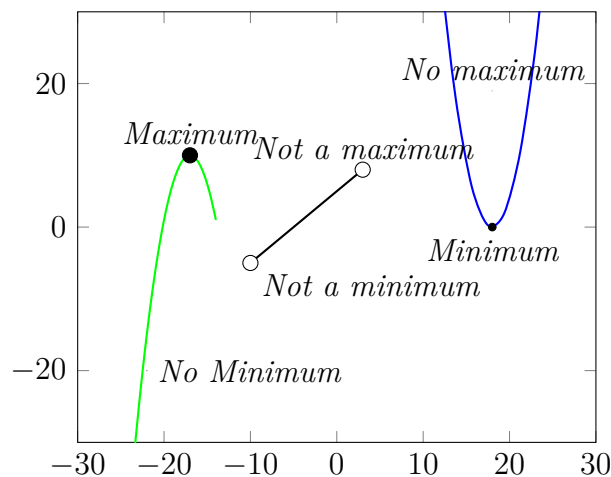
Definition 1. Let f be defined on an interval I containing c .

1. $f(c)$ is the **minimum** of f on I when $f(c) \leq f(x)$ for all $x \in I$.
2. $f(c)$ is the **maximum** of f on I when $f(c) \geq f(x)$ for all $x \in I$.

The minimum and maximum of a function on an interval are the **extreme values**, or **extrema** (the singular form of extrema is extremum) of the function on the interval. The minimum and maximum of a function on an interval are also the **absolute minimum** and **absolute maximum**, or the **global minimum** and **global maximum**, on the interval. Extrema can occur at interior points or endpoints of the interval. Extrema that occur at the endpoints are called **endpoints extrema**.



Remark 2. A function need not have a minimum or a maximum on an interval.



The proofs of the following theorems are not within the scope of this course. However, they play central role in searching values where maximum or minimum of a function is attained.

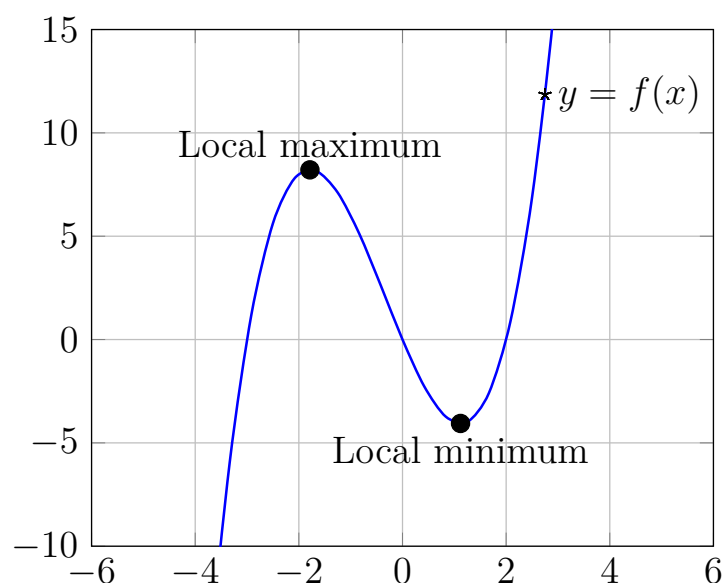
Theorem 3 (INTERMEDIATE VALUE THEOREM). *Let f be a continuous function on the closed bounded interval $[a, b]$. If α is between $f(a)$ and $f(b)$, that is, $(f(a) - \alpha)(f(b) - \alpha) \leq 0$, then there is a value $c \in [a, b]$ such that $\alpha = f(c)$.*

Theorem 4 (THE EXTREME VALUE THEOREM). *If f is continuous on a closed interval $[a, b]$, then f has both a minimum and a maximum on the interval.*

Definition 5. DEFINITION OF RELATIVE EXTREMA

1. If there is an open interval containing c on which $f(c)$ is a maximum, then $f(c)$ is called a **relative maximum** of f or you can say that f has a **relative maximum** at $(c, f(c))$.
2. If there is an open interval containing c on which $f(c)$ is a minimum, then $f(c)$ is called a **relative minimum** of f or you can say that f has a **relative minimum** at $(c, f(c))$.

The plural of relative maximum is **relative maxima**, and the plural of relative minimum is **relative minima**. Relative maximum and relative minimum are sometimes called **local maximum** and **local minimum**, respectively.

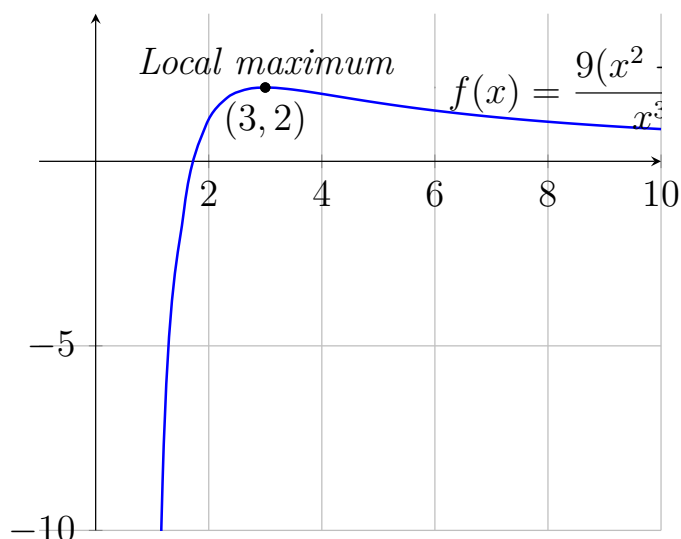


Example 6. The Value of the Derivative at Relative Extrema

Find the value of the derivative at each relative extremum of

$$f(x) = \frac{9(x^2 - 3)}{x^3}$$

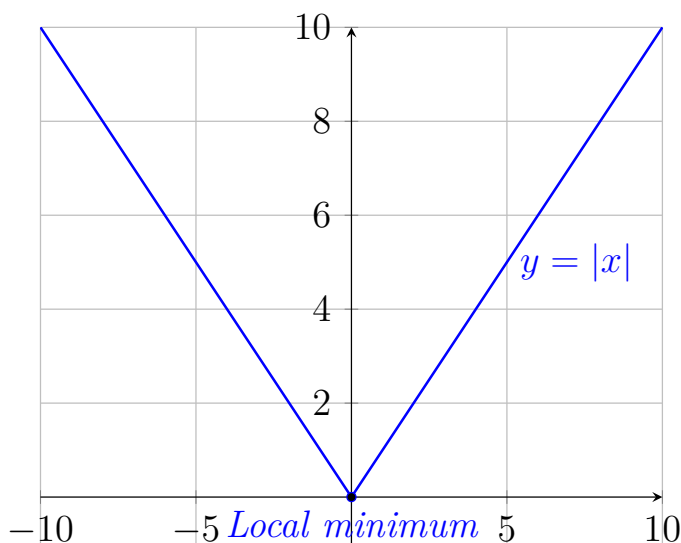
shown in the graph below.



Solution. In class

Example 7. The Value of the Derivative at Relative Extrema

Find the value of the derivative at each relative extremum of $f(x) = |x|$ shown in the graph below.



Solution. In class

Definition 8 (Critical Number). Let f be defined at c . If $f'(c) = 0$ or if f is not differentiable at c , then c is a **critical number** of f .

Theorem 9. If f has a relative minimum or relative maximum at c , then c is a critical number of f .

GUIDELINES FOR FINDING EXTREMA ON A CLOSED INTERVAL.

To find the extrema of a continuous function f on a closed interval $[a, b]$, use these steps.

1. Find the critical numbers of f in (a, b) .
2. Evaluate f at each critical number in (a, b) .

3. Evaluate f at each endpoint of $[a, b]$.
4. The least of these values is the minimum. The greatest is the maximum.

Example 10. *Finding Extrema on a Closed Interval*

- (a) Find the extrema of $f(x) = 3x^4 - 4x^3$ on the interval $[-1, 2]$.
- (b) Find the extrema of $f(x) = 2x - 3x^{\frac{2}{3}}$ on the interval $[-1, 3]$.
- (c) Find the extrema of $f(x) = 2 \sin x - \cos 2x$ on the interval $[0, 2\pi]$.

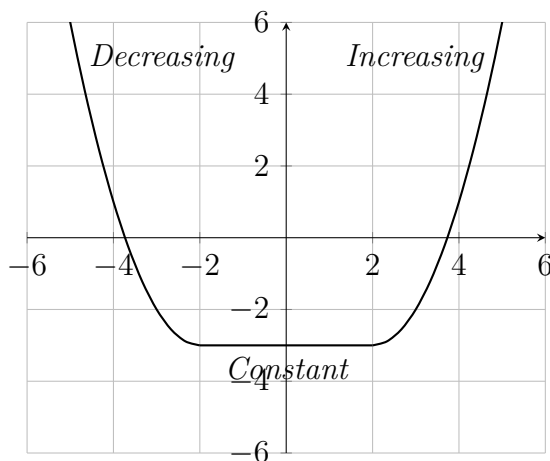
Solution. *In class*

Increasing and Decreasing Functions and The First Derivative Test

Definition 11. *Definitions of Increasing and Decreasing Functions*

1. A function f is **increasing on an interval** when, for any two numbers x_1 and x_2 in the interval, $x_1 < x_2$ implies $f(x_1) < f(x_2)$.
2. A function f is **decreasing on an interval** when, for any two numbers x_1 and x_2 in the interval, $x_1 < x_2$ implies $f(x_1) > f(x_2)$.

A function is increasing when, as x moves to the right, its graph moves up, and is decreasing when its graph moves down.



Theorem 12. *Test for Increasing and Decreasing Functions*

Let f be a function that is continuous on the closed interval $[a, b]$ and differentiable on the open interval (a, b) .

1. If $f'(x) > 0$ for all x in (a, b) , then f is increasing on $[a, b]$.
2. If $f'(x) < 0$ for all x in (a, b) , then f is decreasing on $[a, b]$.
3. If $f'(x) = 0$ for all x in (a, b) , then f is constant on $[a, b]$.

Example 13. Find the open intervals on which

$$f(x) = x^3 - \frac{3}{2}x^2$$

is increasing or decreasing.

Solution. In class

GUIDELINES FOR FINDING INTERVALS ON WHICH A FUNCTION IS INCREASING OR DECREASING

Let f be continuous on the interval (a, b) . To find the open intervals on which f is increasing or decreasing, use the following steps.

1. Locate the critical numbers of f in (a, b) and use these numbers to determine test intervals.
2. Determine the sign of $f'(x)$ at one test value in each of the intervals.
3. Use Theorem 3.5 to determine whether f is increasing or decreasing on each interval.

These guidelines are also valid when the interval (a, b) is replaced by an interval of the form $(-\infty, b)$, (a, ∞) , or $(-\infty, \infty)$.

Definition 14. A function f is said to be **strictly monotonic** on an interval when it is either increasing on the entire interval or decreasing on the entire interval.

Example 15. The function $f(x) = x^3$ is strictly monotonic on $(-\infty, \infty)$.

Theorem 16 (THE FIRST DERIVATIVE TEST). Let c be a critical number of a function f that is continuous on an open interval I containing c . If f is differentiable on the interval I , except possibly at c , then $f(c)$ can be classified as follows.

1. If $f'(x)$ changes from **negative to positive** at c , then f has a **relative minimum** at $(c, f(c))$.
2. If $f'(x)$ changes from **positive to negative** at c , then f has a **relative maximum** at $(c, f(c))$.
3. If $f'(x)$ is positive on both sides of c or negative on both sides of c , then $(c, f(c))$ is neither a relative minimum nor a relative maximum.

Example 17. Applying the First Derivative Test, find the relative extrema of

$$f(x) = \frac{1}{2}x - \sin x$$

on the interval $(0, 2\pi)$.

Solution. In class

Example 18. Applying the First Derivative Test, find the relative extrema of $f(x) = (x^2 - 4)^{\frac{2}{3}}$.

Solution. In class

Example 19. Applying the First Derivative Test, find the relative extrema of

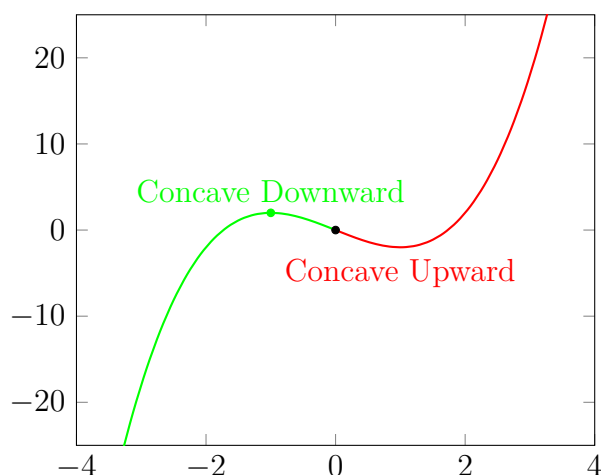
$$f(x) = \frac{x^4 + 1}{x^2}.$$

Solution. In class

Concavity and the Second Derivative Test

Concave describes an **inward curve**, and is used to describe gentle, subtle curves, like the kinds found in mirrors or lenses.

Definition 20. Let f be differentiable on an open interval I . The graph of f is **concave upward** on I when f' is increasing on the interval, and **concave downward** on I when f' is decreasing on the interval.



Remark 21. Concavity is not defined for a line. A straight line is **neither concave upward nor concave downward**.

The following graphical interpretation of concavity is useful.

1. Let f be differentiable on an open interval I . If the graph of f is concave upward on I , then the graph of f lies above all of its tangent lines on I .
2. Let f be differentiable on an open interval I . If the graph of f is concave downward on I , then the graph of f lies below all of its tangent lines on I .

Theorem 22 (The Second Derivative Test for Concavity). Let f be a function whose second derivative exists on an open interval I .

1. If $f''(x) > 0$ for all x in I , then the graph of f is **concave upward** on I .
2. If $f''(x) < 0$ for all x in I , then the graph of f is **concave downward** on I .

To apply the above Theorem, locate the x -values at which $f''(x) = 0$ or f'' does not exist. Use these x -values to determine test intervals. Finally, test the sign of f'' in each of the test intervals.

Example 23. Determine the open intervals on which the graph of $f(x) = \frac{6}{x^2 + 3}$ is concave upward or downward.

Solution. In class

Remark 24. When there are x -values at which the function f is not continuous, these values should be used, along with the points at which $f''(x) = 0$ or f'' does not exist, to form the test intervals.

Example 25. Determine the open intervals on which the graph of $f(x) = \frac{x^2 + 1}{x^2 - 4}$ is concave upward or concave downward.

Solution. In class

Definition 26 (Definition of Point of Inflection). Let f be a function that is continuous on an open interval, and let c be a point in the interval. If the graph of f has a tangent line at this point $(c, f(c))$, then this point is a **point of inflection** of the graph of f when the concavity of f changes from upward to downward (or downward to upward) at the point.

Theorem 27 (Points of Inflection). If $(c, f(c))$ is a point of inflection of the graph of f , then either $f''(c) = 0$ or f'' does not exist at $x = c$.

Example 28. Determine the points of inflection and discuss the concavity of the graph of $f(x) = x^4 - 4x^3$.

Solution In class

Theorem 29 (Second Derivative Test). Let f be a function such that $f'(c) = 0$ and the second derivative of f exists on an open interval containing c .

1. If $f''(c) > 0$, then f has a relative minimum at $(c, f(c))$.
2. If $f''(c) < 0$, then f has a relative maximum at $(c, f(c))$.

If $f''(c) = 0$, then the test fails. That is, f may have a relative maximum, a relative minimum, or neither. In such cases, you can use the First Derivative Test

Example 30. Find the relative extrema of $f(x) = -3x^5 + 5x^3$.

Solution. In class